demonstrating that unmanned systems could engage targets with high accuracy and minimal collateral damage.

- *Enhanced sensor and communication systems*: The introduction of synthetic aperture radar (SAR), EO/IR imaging, and real-time SATCOM links enabled high-fidelity intelligence gathering and long-distance command-and-control.
- *Autonomy and system redundancy*: Advances in autopilot algorithms, navigation systems, and onboard data processing improved UAV reliability, enabling semi-autonomous operation in multi-threat environments.
- *Global proliferation and standardization*: During this period, UAV technology expanded internationally, with major contributions from the United States, Israel, France, the United Kingdom, and China, reflecting a global standardization of unmanned systems for both military and civilian use.

4.5.2 Key pioneers and notable drone models

4.5.2.1 RQ-1 Predator (1995)

The RQ-1 Predator, developed by General Atomics Aeronautical Systems (GA-ASI) for the United States Air Force (USAF) and Central Intelligence Agency (CIA), represents one of the most transformative UAVs in aviation history and a defining model of the third generation of drones. First introduced in 1995, the Predator revolutionized modern warfare by merging realtime surveillance and precision strike capabilities within a single, remotely operated platform. Designed as a medium-altitude long-endurance (MALE) drone, the RQ-1 Predator established the operational framework for future unmanned systems by combining endurance, intelligence-gathering precision, and armed engagement capacity (Amburn, et al. 2005).